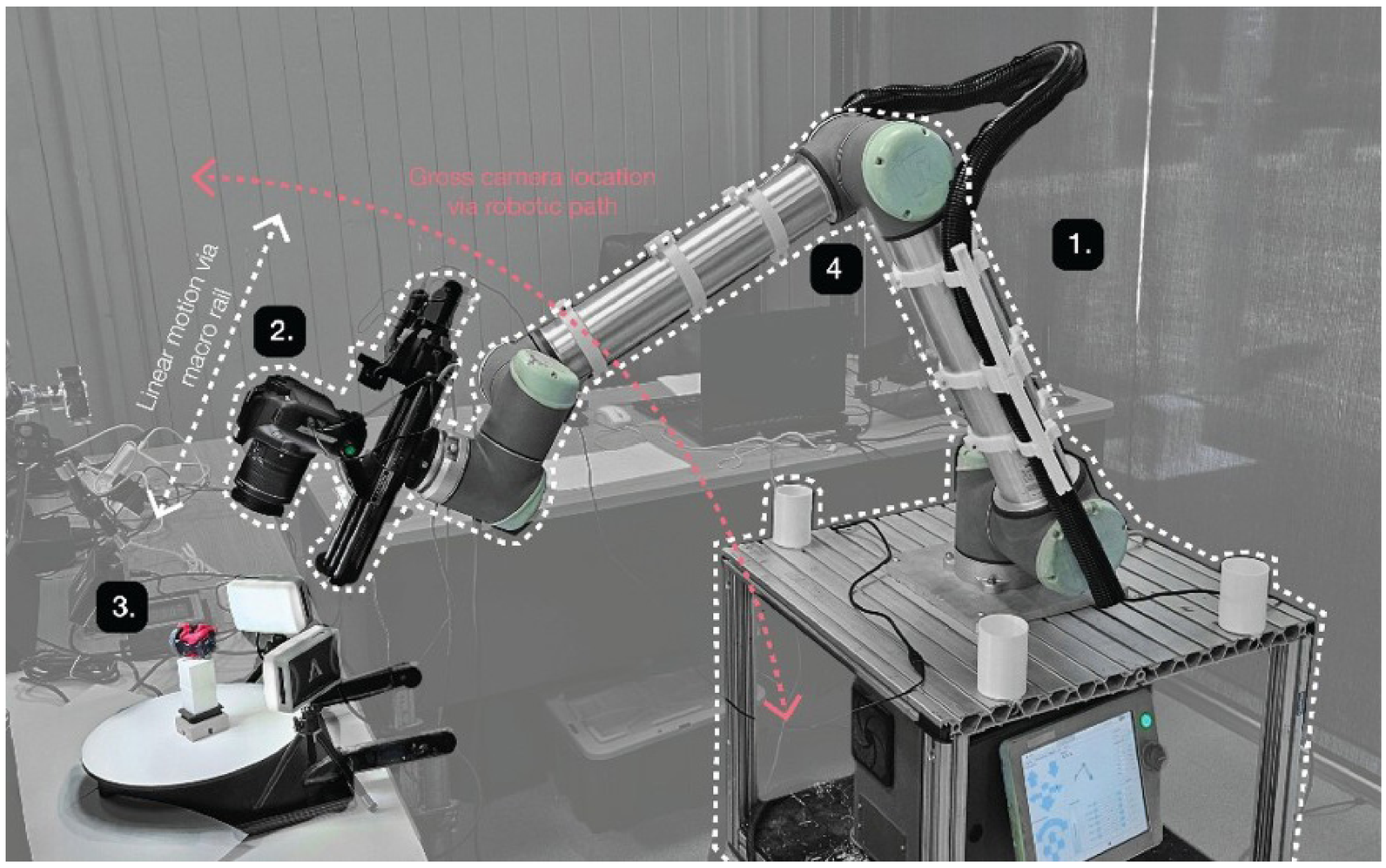
Redistributing Human Work in Robotic-Assisted Photogrammetry and Focus Stacking for Small Museum Object Digitisation

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Context & Motivation

High-resolution digitisation of small museum artefacts requires careful coordination across image acquisition, processing, and reconstruction. While robotic positioning and focus stacking support repeatable multi-view capture, integrating these technologies does not necessarily eliminate human involvement.

Research Questions

1. How does robotic assistance redistribute human work across small-object digitisation workflows?
2. Where do interaction challenges emerge in calibration, handoff, monitoring, data management, and output assessment?
3. What design considerations can inform future robotic-assisted digitisation systems that make these workflows more legible, recoverable, and usable in practice?

DIGITISATION WORKFLOW

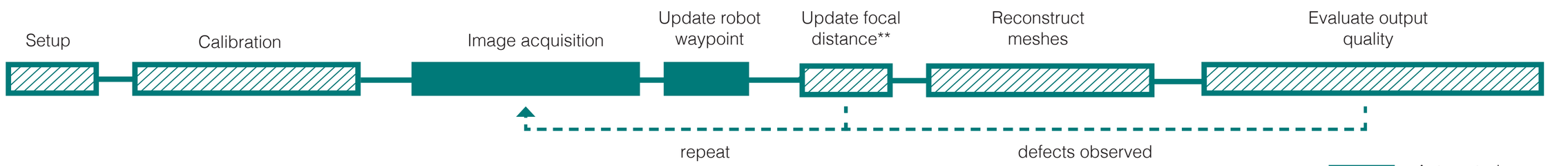
METHOD: Four researcher-operated sessions were analysed using field notes, observational records, process images, acquisition timings and expert output review. The workflows used different artefacts and capture settings; comparisons are descriptive.

A — PLAN + CALIBRATE
Register the turntable, camera and robot; check framing and collision clearance.

B — CAPTURE + REPOSITION
Run multi-view capture; adjust focal range and manually reorient objects.

C — PROCESS + RECONSTRUCT
Organise image sets, generate focal stacks, mask and reconstruct meshes.

D — ASSESS THE OUTPUT
Inspect coverage, surface detail and defects against the intended task.



CASE STUDIES (4 SESSIONS, 3 ARTEFACTS)



A ‘Sarcophagus’ (approx. 45mm)
Complex geometry and varied surface conditions.



B ‘Pot’ (approx. 42mm)
Curved geometry and fine surface detail.



C Red Dragon figurine (approx. 40mm)
Intricate details and occluded areas.

KEY FINDINGS

Calibration
Calibration errors could affect many images before becoming apparent during review.

Redistributed Work
Human work shifted toward preparation, supervision, coordination and recovery.

Device Handoffs
Operators coordinated transitions between devices and checked when capture could continue.

Quality Judgement
Model adequacy depended on the documentation purpose.

IMPLICATIONS FOR HCI

Make calibration and device handoffs visible and recoverable.

Human roles shift to planning, judgement and quality control

Design for flexible setups to accommodate various artefacts

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